# Migrating to Firestore

A Comprehensive Guide to Transitioning from Self-Hosted MongoDB to Serverless MongoDB Compatibility (Enterprise Edition GA August 2025)

# Native vs. Compatibility

Understanding the two modes of Firestore operation

# FLEXIBLE OPERATING MODES

## Firestore Native

Proprietary SDKs and query language. Optimized for real-time listeners and seamless integration with Firebase Auth, Functions, and Storage.

Best for new greenfield projects requiring deep Firebase ecosystem integration.

## MongoDB Compatibility

Enterprise Edition feature using standard MongoDB drivers, tools (mongosh, Compass), and libraries with zero code changes.

Best for migrating existing MongoDB workloads to serverless infrastructure.

# The Serverless Advantage

Why move from Self-Hosted to Fully Managed?

# UNLOCKING OPERATIONAL EFFICIENCY

## Zero Management

No more managing replica sets, sharding, or patching. Google handles the entire infrastructure lifecycle.

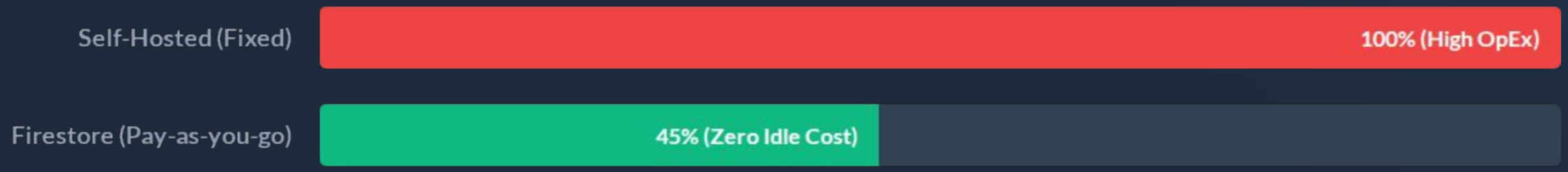
## Elastic Scaling

Virtually unlimited horizontal scaling handles traffic spikes automatically without manual intervention.

## High Availability

99.999% SLA with multi-region replication and strong consistency out of the box.

# COST COMPARISON (TCO)




*Self-hosted infrastructure carries constant costs for idle resources and database administration, whereas Firestore scales cost linearly with actual usage.*

# Migration Pathways

Strategies for different project stages

# NEW VS. EXISTING PROJECTS

## Empty Projects

Connect directly via the connection string. No transformation needed. All new writes go to serverless storage immediately.

## Historical Data

Utilize batch scripts for small datasets or Google Datastream for zero-downtime, continuous replication of large databases.

# MIGRATION TOOLING ECOSYSTEM

| Tool / Method | Recommended Use Case | Complexity |
|---|---|---|
| Custom Batch Script | Small to Moderate (< 500GB) data cleanup & migration | Low (Python/Java) |
| Google Datastream | Zero-downtime migration for large databases | High (CDC Based) |
| Cloud Dataflow | Bulk loading and large-scale transformation | Medium (Managed) |
| Direct Connection | Brand new development or small manual inserts | Zero |

# Constraints & Differences

Critical technical considerations for a smooth transition

# TECHNICAL GUARDRAILS

- **Document Size:** Max 4 MiB (compared to MongoDB's 16 MiB). Offload large files to Cloud Storage.
- **Nesting Depth:** Maximum of 20 levels deep for nested objects or arrays.
- **Naming Rules:** No `$` characters, no leading `__` in field names, and no empty collection names.
- **BSON Support:** Decimal128 and certain Timestamp variants may require conversion before migration.

# Q&A

Ready to transition to a modern serverless architecture?

**Step 1:** Proof of Concept

**Step 2:** Scan Data for Limits

**Step 3:** Execute Batch Migration

# IMAGE SOURCES

https://miro.medium.com/v2/resize:fit:520/1*7nTsfZaGccJFKI8t7YyuPQ.png

Source: medium.com

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https://slideuplift.com/wp-content/uploads/edd/2020/08/ItemID-1534-Bar-Graph-Cost-Drivers-PowerPoint-Template-1.jpg

Source: slideuplift.com

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https://moldstud.com/uploads/images/database-developer-vs-database-administrator-understanding-the-difference.webp

Source: moldstud.com

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https://assets.qlik.com/image/upload/w_688/q_auto/qlik/glossary/data-migration/seo-hero-data-migration_igyweg.png

Source: www.qlik.com